STATIONERY SHOP MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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BACHELOR OF ENGINEERING

in

COMPUTER ENGINEERING

GOVERNMENT ENGINEERING COLLEGE
PALANPUR





Gujarat Technological University, Ahmedabad

(Academic Year 2026–2027)





GOVERNMENT ENGINEERING COLLEGE, PALANPUR

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **Stationery Shop Management System Report** has been carried out by **Pratham Parmar** under my guidance in partial fulfillment for the degree of Bachelor of Engineering in Computer Engineering, 7th Semester of Gujarat Technological University, Ahmadabad during the academic year 2026-27.

Sign :

Dr. R. H. Jaiswal

Internal Guide

Sign :

Dr. R. H. Jaiswal

Head of the Department

PROJECT COMPLETION CERTIFICATE

Date: 22/07/2026

To,
Pratham Dipakkumar Parmar

Experience Letter

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Pratham Dipakkumar Parmar** successfully completed an internship with Samudra Technologies Pvt. Ltd. as an **Jr. Software Developer** from **3rd July 2026** to **17th July 2026**.

During the internship, **Pratham Dipakkumar Parmar** was involved in assigned projects and tasks related to the department and demonstrated a sincere willingness to learn, good teamwork, and a professional attitude throughout the internship period. The intern performed the assigned responsibilities with dedication and completed the internship satisfactorily.

We appreciate the efforts and contribution made during the internship and wish Prathamkumar Dipakkumar Parmar every success in future academic and professional endeavors.

We thank them for their association with Samudra Technologies Pvt. Ltd.

For Samudra Technologies Private Limited

Mr. Parth Barot

Director

Samudra Technologies Pvt Ltd

This is a computer-generated document and does not require a physical signature. Digitally signed by the authorized representative.





GOVERNMENT ENGINEERING COLLEGE, PALANPUR

DECLARATION

I hereby declare that the Internship / Project report submitted along with the Internship / Project entitled **"Stationery Shop Management System"** submitted in partial fulfillment for the degree of Bachelor of Engineering in **Computer Engineering** to Gujarat Technological University, Ahmedabad, is a bonafide record of original project work carried out by me at **Samudra Technologies Private Limited** under the supervision of **Dr. R. H. Jaiswal** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of student: **Sign:**

Pratham Parmar (230610107027**)**

ACKNOWLEDGEMENT

I express my sincere gratitude to Government Engineering College, Palanpur and Shree Sales Agency for providing me with the opportunity to undertake and execute my software engineering project, **Stationery Shop Management System**. I am especially thankful to **Dr. R. H. Jaiswal**, Head of the Computer Engineering Department, for continuous guidance, technical insights, and constant encouragement throughout the design and development phases. Furthermore, I am deeply indebted to our domain faculty members for their patient mentorship, technical clarity, and guidance in architectural choices including offline-first SQLite synchronization, GST tax calculation engines, and cross-platform mobile deployment.

I would like to express my heartfelt gratitude to **Dr. K.B. Judal**, Principal of Government Engineering College, Palanpur, for providing an excellent academic environment, laboratory infrastructure, and necessary resources to successfully complete this project.

I am deeply thankful to **Dr. R. H. Jaiswal**, Head of the Department of Computer Engineering, for the constant encouragement and support extended during the project timeline.

I also extend my sincere thanks to the Project Coordinators and Review Committee members for their valuable feedback, timely suggestions, and constructive criticism during milestone evaluations.

I express my gratitude to all the faculty members and technical staff of the Computer Engineering Department, Government Engineering College, Palanpur, for their cooperation and assistance during the system implementation.

Finally, I would like to thank my parents, family members, and peers for their constant motivation, encouragement, and unwavering support, which helped me successfully complete this software engineering endeavor.

I sincerely acknowledge everyone who directly or indirectly contributed to the successful completion of my project and made this learning experience meaningful and rewarding.

PRATHAM PARMAR

Enrollment No.: 230610107027

ABSTRACT

This project report presents the design and implementation of the Stationery Shop Management System, a robust, cross-platform, offline-first software application engineered for point-of-sale (POS) operations, inventory tracking, GST tax invoicing, and sales analytics. Developed using Ionic 8, Angular 20 (standalone components), and Capacitor 8 on the frontend, alongside a Node.js, Express, and MySQL 8 backend, the application functions seamlessly as both a progressive web application and a native Android application.

A key innovation of the system is its offline-first architecture powered by SQLite local caching (@capacitor-community/sqlite on Android and jeep-sqlite WebAssembly over IndexedDB on the web). Read operations are served instantaneously from local storage, while all write transactions—such as bill creation, stock adjustments, and payment processing—are recorded locally first and enqueued into a transactional synchronization queue (sync_queue). A background synchronization engine automatically replays queued operations to the central backend server once network connectivity is restored, assigning sequential 8-digit invoice numbers.

Additional core features include integrated QR and barcode camera scanning via @capacitor-mlkit/barcode-scanning, line-item percentage or fixed-amount discounts, multi-mode payment tracking (Cash, UPI, and Cheque), an Indian GST tax computation engine supporting intra-state (SGST/CGST) and inter-state (IGST) calculations with number-to-words conversion, automated A4 PDF invoice generation using pdfmake, and interactive sales reporting dashboards.

Overall, the system bridges academic software engineering principles with commercial production requirements, delivering zero-downtime retail operations, automated tax compliance, and reliable data synchronization across distributed desktop and mobile clients.

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Full Form / Description
POS	Point of Sale
GST	Goods and Services Tax
SGST	State Goods and Services Tax
CGST	Central Goods and Services Tax
IGST	Integrated Goods and Services Tax
HSN	Harmonized System of Nomenclature
SKU	Stock Keeping Unit
WASM	WebAssembly
PWA	Progressive Web Application
APK	Android Package Kit

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CHAPTER 1: INTRODUCTION

1.1 Background of Point of Sale & Inventory Systems

In today's fast-paced retail and commercial environment, stationery businesses handle hundreds of unique inventory items (SKUs), varying unit pricing, fast-moving consumer transactions, and strict tax compliance regulations. Traditional paper-based billing and manual stock registers are highly error-prone, labor-intensive, and susceptible to discrepancies during peak operational hours. Furthermore, modern micro-enterprises and retail outlets require cross-platform point-of-sale (POS) solutions that run reliably on both mobile devices (smartphones, tablets) and desktop web browsers.

A significant engineering challenge facing modern retail applications is network reliance. Traditional cloud-first billing applications become unusable during internet outages, leading to complete operational downtime, lost sales, and customer dissatisfaction. To address this critical vulnerability, modern software architectures emphasize offline-first design patterns, wherein application data is cached and manipulated locally before asynchronously synchronizing with a centralized cloud backend.

Additionally, commercial software operating in India must comply with the Goods and Services Tax (GST) framework, requiring automated calculation of SGST, CGST (for intra-state sales) and IGST (for inter-state sales), HSN code mapping, precise rounding mechanisms, and standardized A4 invoice generation. Therefore, building an integrated stationery shop management system involves combining mobile engineering, local storage engines, cloud database synchronization, and automated tax accounting.

Through modern hybrid frameworks such as Ionic 8, Angular 20, and Capacitor 8, developers can build a single codebase that delivers a high-performance web app and a native Android application, lowering maintenance overhead while maximizing operational accessibility for shop owners.

Thus, the Stationery Shop Management System represents a comprehensive technical solution engineered to solve real-world retail challenges through offline reliability, high-speed camera barcode scanning, dynamic tax computation, and automated client-server data synchronization.

1.2 Purpose of the Offline-First Stationery Shop Management Application

The primary purpose of the **Stationery Shop Management Application** is to provide a zero-downtime, high-efficiency point-of-sale and inventory system designed explicitly for stationery store operations. Built using modern front-end technology (Ionic 8 + Angular 20 + Capacitor 8) and an express Node.js / MySQL backend, the application guarantees uninterrupted store operations even during severe network interruptions.

Key business objectives of the system include:

- Ensuring continuous billing capability using an embedded SQLite database (`@capacitor-community/sqlite` on mobile devices and `jeep-sqlite` WASM on desktop web browsers).
- Streamlining cart creation via camera-based QR and barcode scanning utilizing Capacitor MLKit, alongside quick category filtering and godown storage location matching.
- Providing flexible discount handling (per-line percentage or fixed rupee amounts) and multi-channel payment flows (Cash, UPI, Cheque with cheque numbers).
- Delivering compliant Indian GST tax invoicing, calculating net values, SGST, CGST, IGST, nearest rupee rounding, and automated Indian number-to-words conversion (Lakhs and Crores).
- Enabling automatic transaction queueing (`sync_queue`) and background replay to assign official 8-digit sequential invoice numbers upon cloud connection.

By unifying mobile agility with robust backend synchronization, the project provides shop owners with a dependable digital asset that increases billing speed and simplifies statutory tax compliance.

1.3 Scope and Technical Objectives

The scope of this project encompasses full-stack software development, offline caching mechanics, tax logic implementation, and native cross-platform build pipelines. The system is designed to fulfill several explicit technical objectives:

Point of Sale (POS) Billing Engine
Implement real-time cart manipulation, per-line fixed/percentage discounts, direct numeric quantity inputs, payment state management (Paid, Pending, Partial), and multi-payment option tracking (Cash, UPI, Cheque).

Inventory and Stock Management
Provide complete CRUD controls for items and categories, low-stock visual warning badges, SKU mapping, per-item HSN codes, configurable GST rates (0%, 5%, 12%, 18%, 28%), and godown location search matching.

GST Tax Calculation & PDF Invoicing
Build a dynamic GST computation utility (`gst.util.ts`) that handles intra-state vs inter-state tax rules based on seller/buyer state codes (defaulting to Gujarat 24), outputs structured A4 PDF tax invoices via `pdfmake`, and converts totals to Indian currency words.

Offline Caching and Queue Synchronization
Develop a universal `SqliteService` interface selecting native SQLite on Android and WebAssembly `jeep-sqlite` on browsers. All writes generate local database rows and enqueue backend requests in `sync_queue` for linear background replaying upon network reconnection.

Analytics and Reporting Dashboard
Deliver visual sales intelligence including daily sales charts, top-selling product lists, low-selling items, and low-stock warnings with 7-day and 30-day temporal toggles.

CHAPTER 2: COMPANY PROFILE & MENTORSHIP FRAMEWORK

2.1 Company Profile & Corporate Overview

Samudra Technologies (**Samudra Technologies Private Limited**) is a premier technology solutions provider specializing in Dairy Industry Automation and Enterprise Software Engineering within the Information Technology sector.

Corporate Parameter	Details & Specifications
Company Name	Samudra Technologies Private Limited
Legal Entity & CIN	Private Limited Company (CIN: U28251GJ2023PTC145536)
Date of Incorporation	October 17, 2023 (ROC Ahmedabad)
Industry Sector	Information Technology — Dairy Industry Automation
Quality Certification	ISO 9001:2000 Certified
Official Website	www.samudratech.com
Registered Office	S.F 210, I Square, Near Shukan Mall Cross Road, Sola, Ahmedabad, Gujarat – 380060
Regional Branch Office	3rd Floor, Samudra, Ambica Nagar Road, Suresh Mehta Chowk, Banaskantha, Gujarat – 385001
Key Leadership	Mr. Pankajkumar Hargovinddas Barot & Mr. Parth Barot (Director)

Table 2.1: Samudra Technologies Corporate Profile Overview

2.2 Background, History & Industry Positioning

Samudra Technologies has established a strong market presence in dairy industry automation, leveraging close to 25 years of domain expertise in IT-driven dairy solutions. Operating as a key player in technology integration and consulting, the company delivers end-to-end services through the broader network of **SIPL Group** companies, serving regional, national, and international clients.

2.3 Vision, Mission & Strategic Focus

The corporate mission of Samudra Technologies centers on acquiring deep context into customer needs and engineering custom, high-impact software solutions rather than generic off-the-shelf products. Guided by the core tagline **"Empowering Farmers, Enabling Cooperatives, Transforming Dairy,"** the company focuses on digitalizing agricultural and co-operative workflows.

2.4 Product Portfolio & Flagship Offerings

Samudra Technologies delivers a comprehensive ecosystem of hardware and software products:

- **Dairy ERP Solutions:** End-to-end enterprise resource planning tailored specifically for complex dairy supply chains and processing plants.
- **Milk Procurement Systems (MPS):** Advanced procurement software, highlighted by the flagship **Smart MPS** platform deployed at Village Level Milk Collection Centres. Built on Android technology, Smart MPS integrates electronic weighing scales, milk analyzers, thermal printers, digital displays, and UPS systems with automatic cloud synchronization and ledger generation (shift reports, monthly summaries, customer ledgers, and rate tables).
- **IoT-Enabled Smart Devices:** Edge computing hardware for automated data collection at collection points.
- **Mobile Applications:** Cross-platform Android/iOS applications for field officers, dairy management, and collection point operators.
- **Industrial & Cooperative Automation:** Automated liquid milk processing, chilling center data logging, and society billing software.
- **Business Intelligence Platforms:** Data analytics and reporting dashboards for data-driven dairy decision-making.

2.5 Target Client Base & Organizational Scale

Samudra Technologies caters to a diverse client ecosystem including Milk Processing Dairies, Co-operative Milk Societies, Credit Co-operative Societies, Educational Institutions/Schools, and Testing Equipment Manufacturers.

- **Financial Details:** Authorized Capital of ₹10.00 Lakh and Paid-up Capital of ₹2.50 Lakh, achieving ₹49.70 Lakh in Revenue for FY ending March 31, 2025.
- **Work Environment:** Glassdoor employee rating of 3.9/5, recognizing the organization's flexible, learning-oriented, and supportive workplace culture.

2.6 Industry Footprint & Mentorship Framework

The company maintains an active industry profile, participating in national events such as the **52nd Dairy Industry Conference & Exhibition 2026** at Yashobhoomi Convention Centre, Delhi. Through active digital channels (YouTube, LinkedIn, Instagram), Samudra Technologies provides training, technical education, and client support.

During the internship period, Samudra Technologies provided a structured technical mentorship environment. Senior leadership and engineering mentors provided guidance in full-stack architecture, offline-first database synchronization, client-side performance optimization, and cross-platform mobile deployment, directly enriching the development of the **Stationery Shop Management System**.

CHAPTER 3: SYSTEM ARCHITECTURE & TECHNICAL STACK

3.1 Frontend Architecture (Ionic 8, Angular 20, Capacitor 8)

The client application is built upon **Ionic 8** and **Angular 20** utilizing Angular's modern Standalone Component paradigm. This architecture eliminates unnecessary NgModule boilerplate, promotes tree-shaking, and improves application startup performance across both mobile browsers and native web views.

Native device bridging is facilitated by **Capacitor 8**. Key Capacitor plugins leveraged in the system include `@capacitor/filesystem` for file management, `@capacitor/preferences` for local key-value configuration (`SellerConfigService`), and `@capacitor-mlkit/barcode-scanning` for hardware-accelerated camera barcode scanning.

Navigation is structured into a five-tab main view (Dashboard, Items, Billing, Reports, Bills) complemented by an expandable side drawer menu (`ion-menu`) providing fast access to Category management, Business/GST Settings, and User Authentication controls.

Additionally, **Claude Code** was utilized as an AI-powered software engineering tool to assist in generating, architecting, and structuring the complete codebase for both the frontend and backend modules of the Stationery Shop Management System.

Architectural Layer	Technology Stack & Core Libraries
Frontend Framework	Ionic 8 & Angular 20 (Standalone Components)
Native Mobile Engine	Capacitor 8 Bridge Engine
Offline Database & WASM	@capacitor-community/sqlite (Android) & jeep-sqlite WASM (Web)
Camera Barcode Scanner	@capacitor-mlkit/barcode-scanning
PDF Invoicing Engine	pdfmake Library
Backend Server & API	Node.js & Express RESTful Framework
Database & Security	MySQL 8 Relational Database & Stateless JWT Authentication

Table 3.1: Full-Stack Technology Framework & Core Libraries

3.2 Backend Architecture & Database (Node.js, Express, MySQL 8)

The centralized backend is implemented as a RESTful web service engineered with **Node.js** and **Express**, connecting to a robust **MySQL 8** relational database.

Backend Architectural Principles

- Stateless JWT Authentication for secure API communication.

- Client-Authoritative Stock tracking: The server respects absolute stock quantities calculated offline by client devices during sync replay.
- Sequential 8-Digit Invoice Generation: Assigns formatted numbers (`00000001`, `00000002`) to synced bills upon database insertion.
- Audit Safe Deletion: Bill deletions execute soft moves to a `deleted_bills` archive table maintaining transactional compliance.
- Idempotent Database Migrations: Seed scripts automate schema creation and admin user provisioning seamlessly.

3.3 Offline-First Synchronization & Cache Strategy

The core innovation of the system is its dual-engine local storage layer abstracted by `SqliteService`. When executing inside native Android containers (`Capacitor.getPlatform() === 'android'`), the application invokes `@capacitor-community/sqlite` directly interacting with native device storage. When running as a website (`ionic serve`), it initializes `jeep-sqlite`, leveraging WebAssembly (WASM) compiled SQLite persisted to browser IndexedDB.

All data read queries (browse items, cart calculation, bill history) query local SQLite tables instantly. Write operations (add bill, edit stock) execute locally and record a JSON mutation paylod into a local `sync_queue` table. `SyncService` monitors network status, running background replays on reconnect, on resume, and every 30 seconds to flush queued actions to the Node.js backend.

CHAPTER 4: ROLES, RESPONSIBILITIES & PROJECT MODULES

4.1 Chronological Breakdown of Development Milestones

The development lifecycle of the Stationery Shop Management Application was executed across three structured engineering phases over a dedicated project development timeline.

Phase I: UI Design, Component Architecture, and Local Storage Setup (Days 1–5)

The initial phase focused on UI layout design using Ionic 8 components, standalone route definitions in Angular 20, local SQLite database schema design (`auth_users`, `items`, `categories`, `bills`, `sync_queue`), and establishing local seed data.

Key accomplishments in Phase I:

- Configured Ionic 8 and Angular 20 route guards and tab layouts.
- Implemented local authentication service with `auth_users` table.
- Configured universal SQLite service picking native or WASM engines.
- Created CRUD UI for Item and Category management.
- Integrated Capacitor Preferences for local seller profile storage.

Phase II: POS Billing Engine, GST Utility, and Camera Scanner (Days 6–10)

The second phase centered on implementing POS cart mechanics, fixed/percentage discounts, barcode scanning integration via MLKit, and building the Indian GST calculation engine.

Key accomplishments in Phase II:

- Engineered `gst.util.ts` for SGST/CGST/IGST tax breakdowns.
- Implemented Indian currency number-to-words conversion algorithm.
- Integrated `@capacitor-mlkit/barcode-scanning` modal with gallery fallback.
- Built dynamic POS cart with editable inline quantities and per-line discounts.
- Created Payment Modal supporting Cash, UPI, and Cheque numbers.

Phase III: Backend Express/MySQL API, Sync Engine, PDF & Android Build (Days 11–15)

The final phase established the Node.js/MySQL backend server, background `sync_queue` replaying engine, `pdfmake` invoice printing, and Android native APK builds.

Key accomplishments in Phase III:

- Developed Node.js/Express REST backend with MySQL 8 schema.

- Implemented `SyncService` background queue drain and server synchronization.
- Automated 8-digit sequential invoice swapping upon successful sync.
- Configured `PdfService` generating bordered A4 GST tax invoices via `pdfmake`.
- Compiled signed Android release APK via Gradle and Temurin JDK 21.

4.2 Key Roles and Technical Responsibilities

Execution of the Stationery Shop Management System required undertaking several distinct engineering roles throughout the software development lifecycle:

Full-Stack Lead Architect

Responsible for overall system architecture, choosing Ionic 8 + Angular 20 standalone structure, designing SQLite relational schema, configuring Node.js REST controllers, and defining MySQL database migration scripts.

- Designed full stack application architecture and REST API endpoints.
- Engineered local SQLite database tables and sync queue structures.
- Configured CORS, JWT security, and client-authoritative stock rules.

Offline Sync & Storage Engineer

Focused on building universal `SqliteService`, handling runtime engine selection (`jeep-sqlite` WASM vs `@capacitor-community/sqlite`), resolving sql-wasm.wasm alignment issues, and building `SyncService` queue management.

- Implemented cross-platform SQLite initialization.
- Engineered transactional `sync_queue` table replaying logic.
- Handled web IndexedDB persistence flushing (`saveToStore`).

GST Compliance & PDF Engine Developer

Constructed pure mathematical utilities for Indian GST calculations (`calcGstBill`), developed Indian currency words conversion (`numberToWordsIndian`), and configured A4 tax invoice templates in `PdfService` using `pdfmake`.

- Authored unit-tested GST computation utilities (`gst.util.ts`).
- Built bordered A4 PDF templates with bank blocks and total ledgers.
- Integrated native Android file sharing via `@capacitor/filesystem`.

Mobile Build & Release Specialist

Managed Capacitor 8 native Android bindings, Gradle build scripts, keystore signing (`stationery-release.jks`), JDK 21 toolchain dependencies, and Android device debugging via ADB.

- Configured Capacitor Android project settings and permissions.

- Resolved Temurin JDK 21 Gradle dependencies.
- Compiled and verified release APK builds on real hardware.

Quality Assurance & Documentation Specialist

Executed complete offline-to-online sync edge case testing, audited legacy bill backward compatibility, validated GST calculations with Jasmines specs (`npm test`), and compiled final technical project reports.

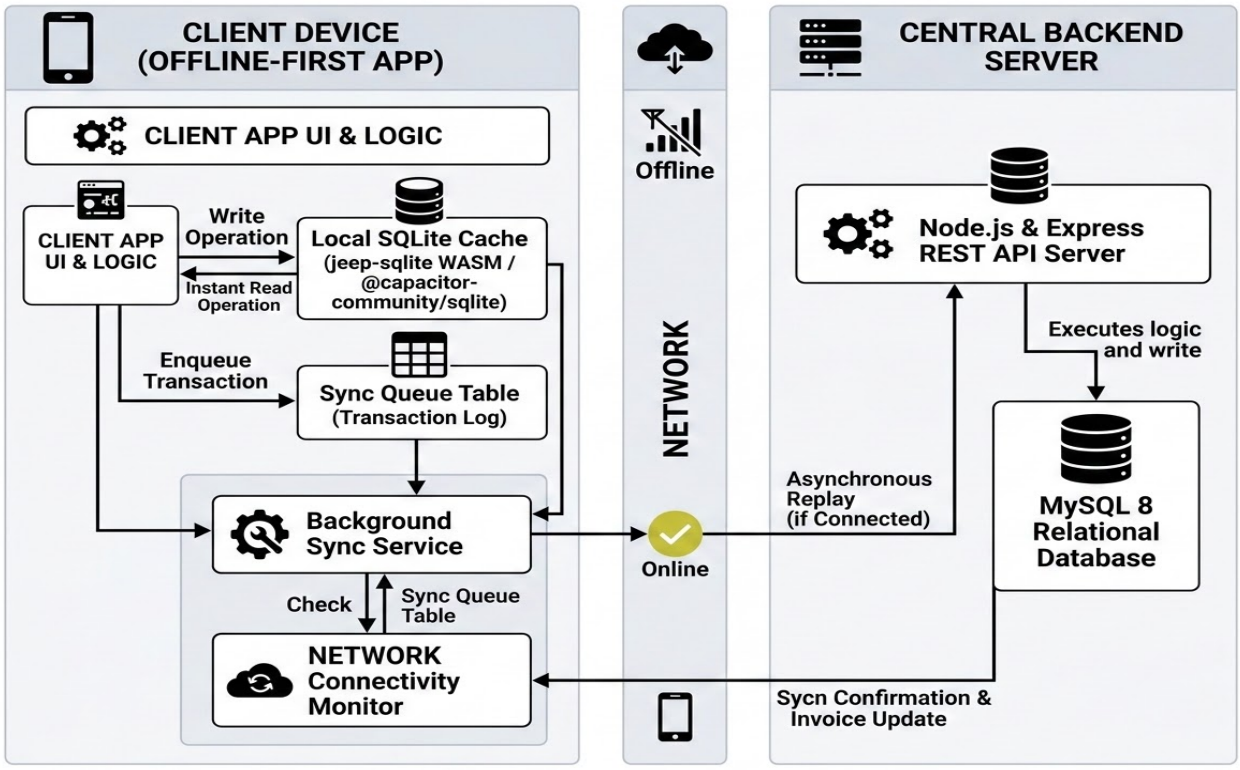
CHAPTER 5: CORE TECHNICAL MODULES & GST ENGINE

5.1 Point of Sale (POS) Billing & Barcode Scanning

The POS Billing module is the primary operational view for store sales staff. It features real-time product search across item names, SKUs, categories, and godown storage locations. Items can be added to cart with instant quantity adjustments (direct numeric typing rather than repeatedly clicking buttons) and per-line percentage or fixed rupee discounts.

To accelerate cart assembly, a camera barcode scanner button is integrated into the billing bar. Powered by `@capacitor-mlkit/barcode-scanning` on native Android devices, tapping the button activates the hardware camera to instantly scan product barcodes/QRs, matching the SKU and placing the item directly into the active cart. On web browsers, the modal transparently falls back to an image gallery picker.

The checkout workflow triggers a payment modal supporting Cash, UPI, and Cheque payment modes (with optional Cheque Number tracking) alongside status tags for Paid, Pending, or Partial payments.



(Fig.5.1 – SQLite Local Storage & Sync Queue Architecture Diagram)

5.2 GST Invoicing & Indian Tax Computation Engine

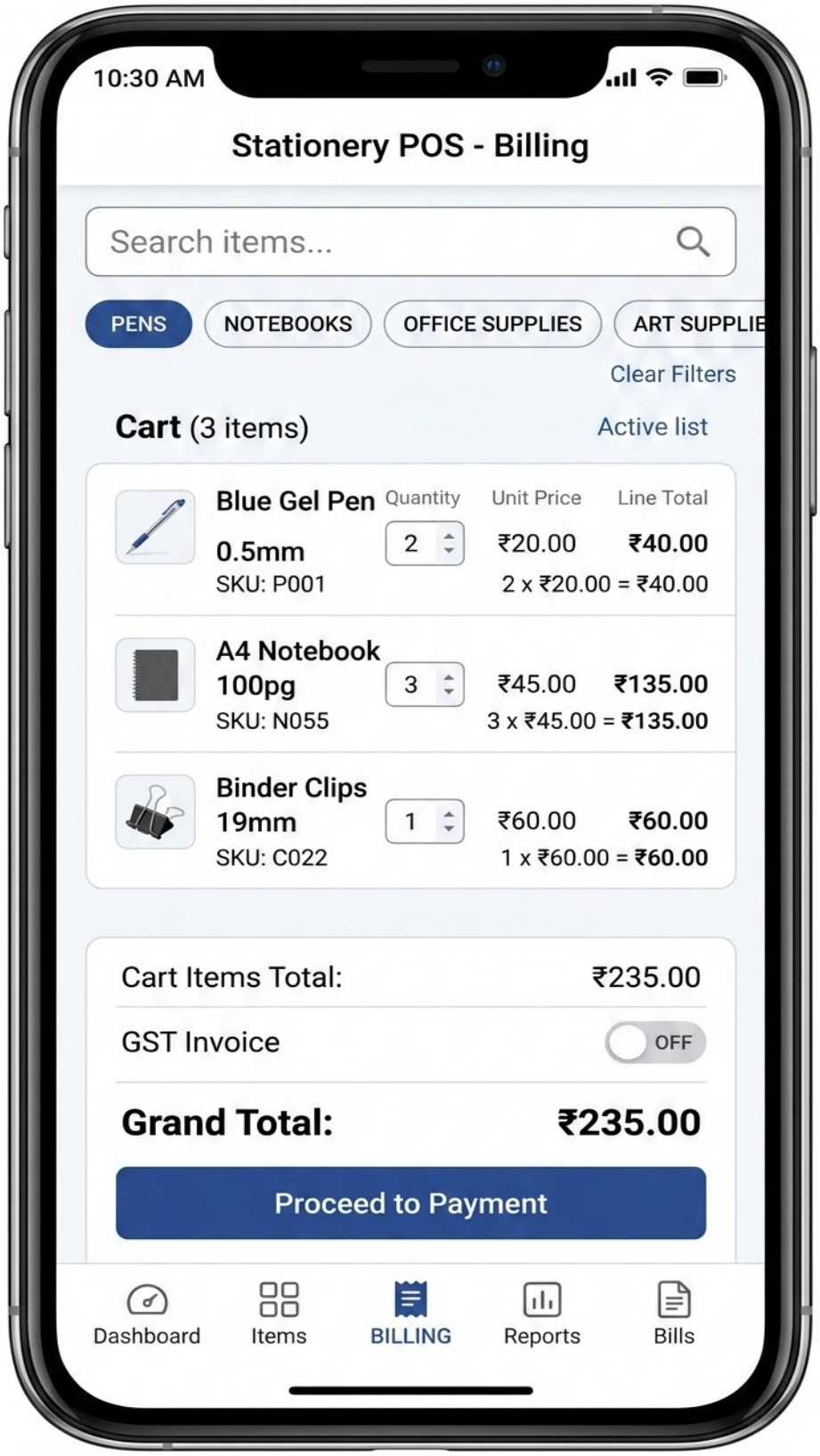
Statutory GST tax compliance is integrated via a dynamic GST Invoice toggle on the billing interface. When activated, the system calculates line-level net taxable values after applying per-line discounts, mapping each item's specific HSN code and GST percentage (0%, 5%, 12%, 18%, 28%).

GST Rate Slab	Statutory Tax Categories
0%	Exempted Goods (Nil Rated, Judicial Paper, Educational Materials)
5%	Essential Student Supplies, Maps, Atlases, Printed Books
12%	Standard Stationery, Notebooks, Exercise Books, Geometry Boxes, Fountain Pens
18%	Office Equipment, Desk Accessories, Calculators, Adhesives, Synthetics
28%	Luxury / Specialty Items, Premium Gift Supplies, High-End Office Products

Table 5.1: Indian GST Rate Slabs & Statutory Tax Categories

The computation engine (`core/utils/gst.util.ts`) compares the seller's state code (configured in Settings via `SellerConfigService`, defaulting to Gujarat / 24) against the buyer's state code. If both match (intra-state), tax is equally split into **SGST** and **CGST**. If they differ (inter-state), tax is calculated as **IGST**.

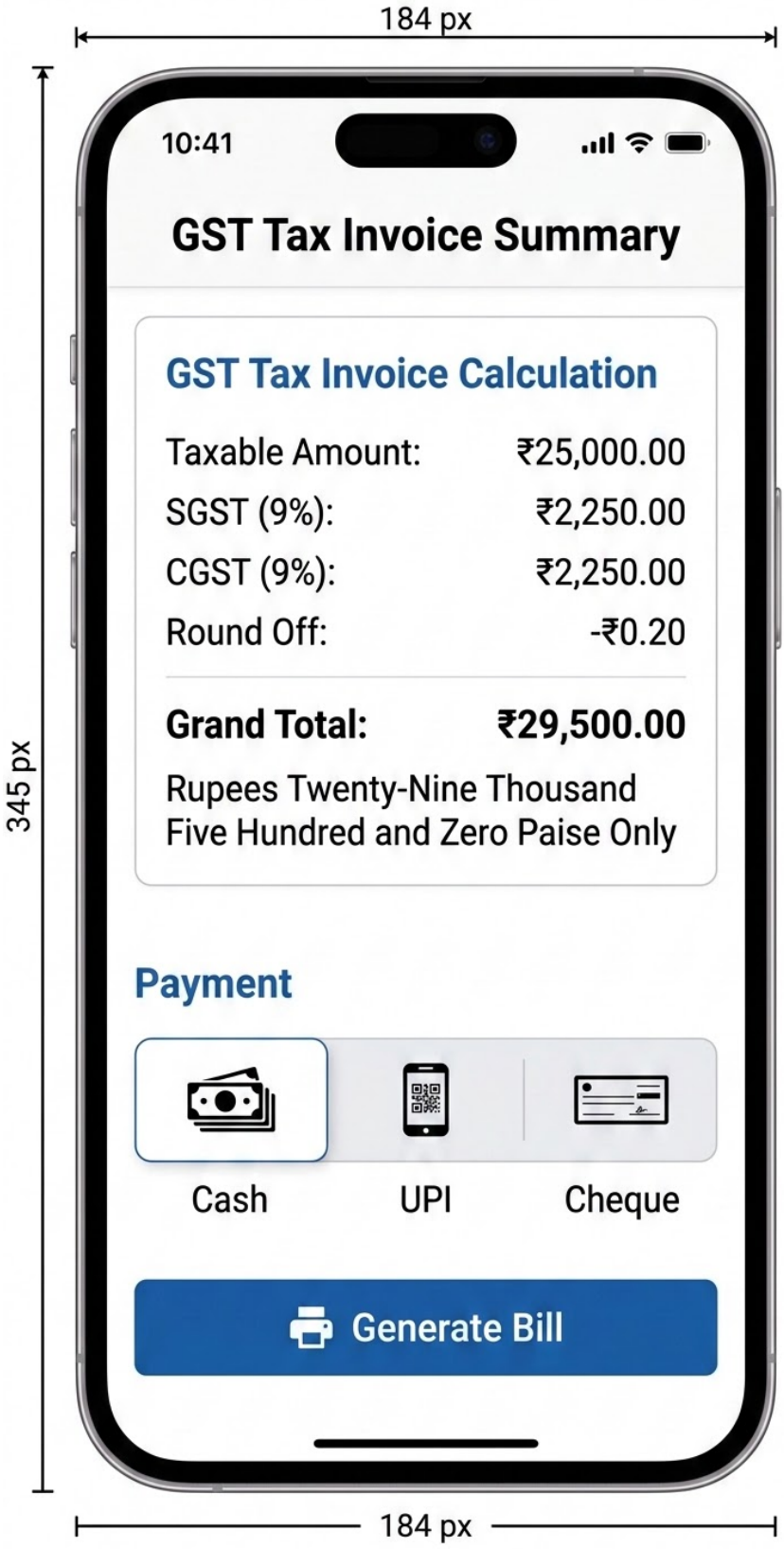
The grand total is rounded to the nearest integer rupee using a signed round-off field. Furthermore, the engine automatically formats the total payable amount into formal Indian English words (using Lakhs and Crores nomenclature, e.g., "Rupees One Lakh Two Thousand Four Hundred Only"). Non-GST bills remain entirely backward compatible, defaulting optional tax fields seamlessly.


10:30 AM
Stationery POS - Billing
Search items...
PENS
NOTEBOOKS
OFFICE SUPPLIES
ART SUPPLIE
Clear Filters
Cart (3 items)
Active list
Blue Gel Pen 0.5mm
SKU: P001
Quantity
Unit Price
Line Total
2
₹20.00
₹40.00
2 x ₹20.00 = ₹40.00
A4 Notebook 100pg
SKU: N055
3
₹45.00
₹135.00
3 x ₹45.00 = ₹135.00
Binder Clips 19mm
SKU: C022
1
₹60.00
₹60.00
1 x ₹60.00 = ₹60.00
Cart Items Total:
₹235.00
GST Invoice
OFF
Grand Total:
₹235.00
Proceed to Payment
Dashboard
Items
BILLING
Reports
Bills

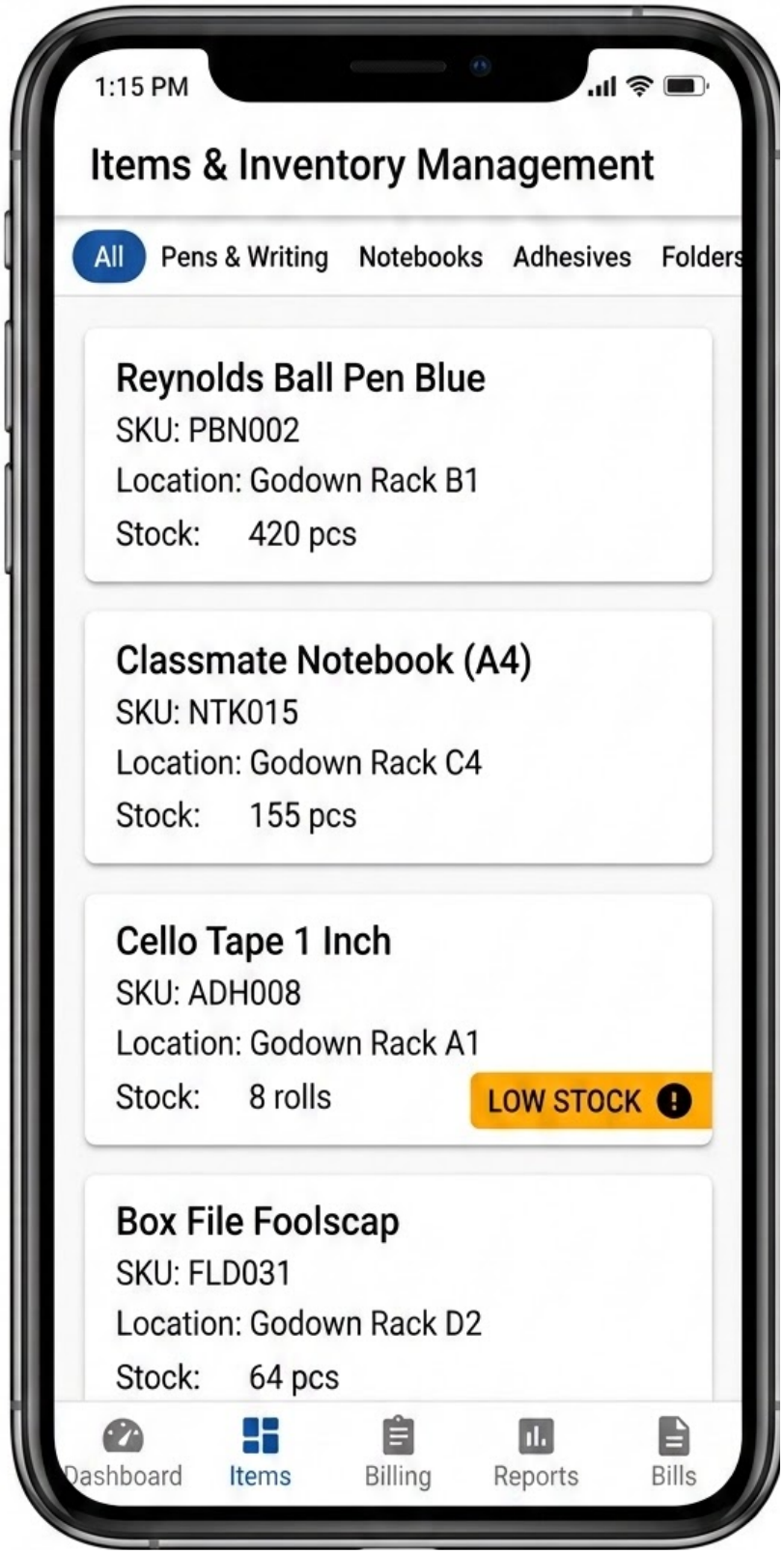
(Fig.5.2 – Point of Sale Billing Screen & Cart Interface)



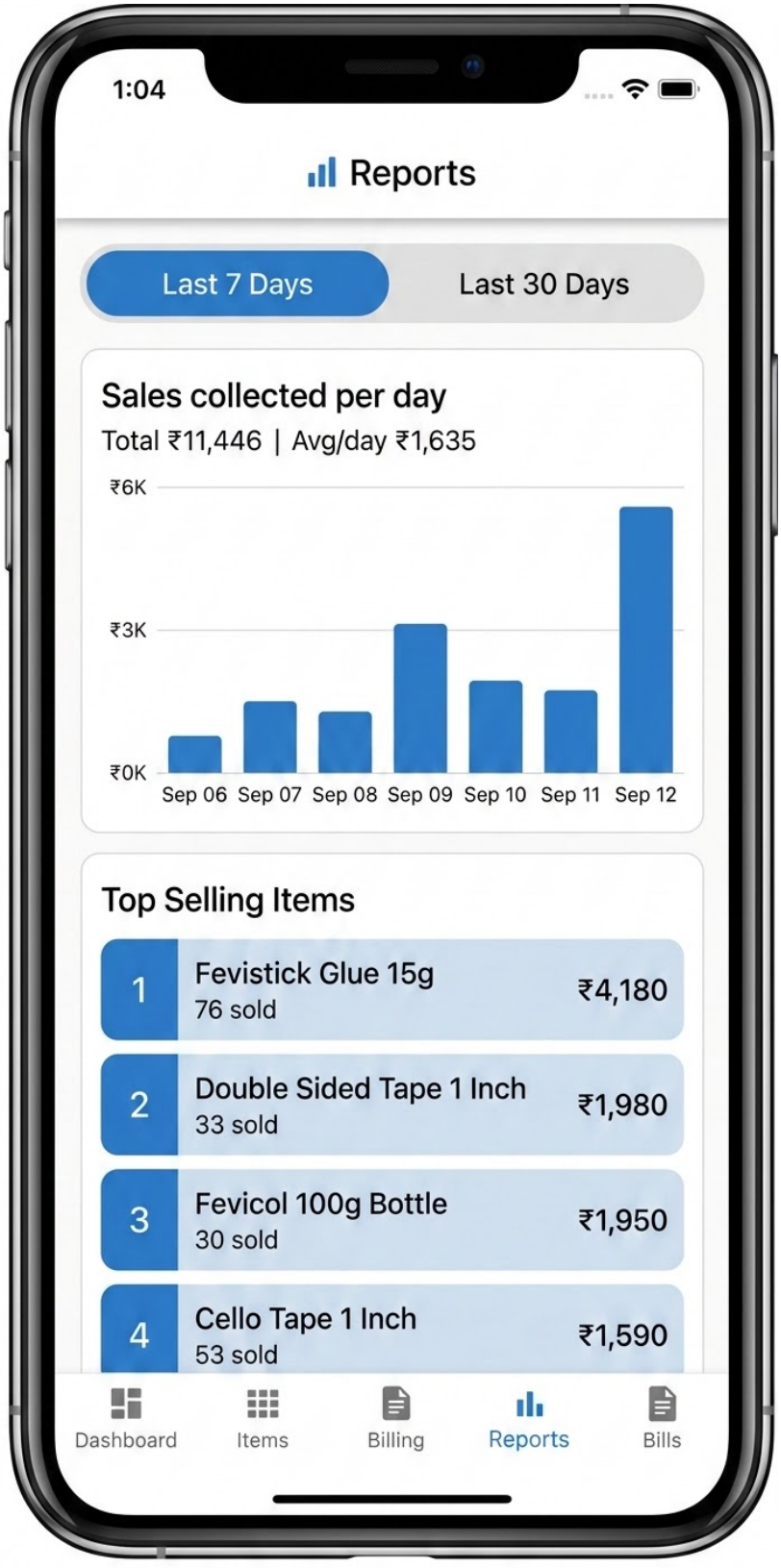
(Fig.5.3 – Barcode / QR Scanner Integration Interface)



(Fig.5.4 – GST Tax Invoice Calculation & PDF View)



(Fig.5.5 – Items & Inventory Management Screen)



(Fig.5.6 – Reports & Daily Sales Analytics Interface)

5.3 Offline Storage Engine (SQLite & jeep-sqlite WASM)

The system utilizes an embedded SQLite database engine for local data persistence. On mobile platforms, native SQLite bindings via Capacitor provide low-latency data operations. On web browsers, the jeep-sqlite library leverages WebAssembly (WASM) backed by IndexedDB persistence to maintain identical database schemas across platforms.

5.4 PDF Generation & Reporting Engine

To maintain inventory integrity and support business intelligence, the application implements structured data management pipelines. Stock items are categorized with distinct SKUs, categories, storage locations, and low-stock threshold triggers (defaulting to 5 units).

When stock falls below threshold limits, visual warning badges appear across item cards and dashboard reporting widgets. The reporting engine aggregates daily sales, top-selling inventory items, and slow-moving products across selectable 7-day and 30-day temporal windows.

Invoice document generation is handled client-side via `PdfService` utilizing `pdfmake`. The service compiles thermal sales receipts or bordered A4 GST Tax Invoices (including HSN summary tables, tax rate ledgers, company bank details, and legal declarations) that can be shared natively on Android via native share sheet or downloaded directly in browser sessions.



(Fig.5.7 – Business Settings & GST Seller Profile Configuration)

CHAPTER 6: SYSTEM DEMONSTRATION & SCREENSHOTS

6.1 Web & Native Application Workspace

The application user workspace provides a responsive, intuitive interface tailored for touch devices and desktop displays. Upon launching, the application checks local SQLite database state, initializing offline caches instantly without blocking user interaction.

An offline status banner (highlighted in yellow) dynamically appears whenever network connectivity drops, accompanied by a cloud synchronization status icon reflecting sync state (Synced, Syncing, N Pending, or Failed). Sales staff can navigate freely between POS Billing, Item Catalog, Sales Reports, and Bill History via the primary bottom tab bar.

6.2 Billing & Inventory Interface Implementation : Stationery POS Module

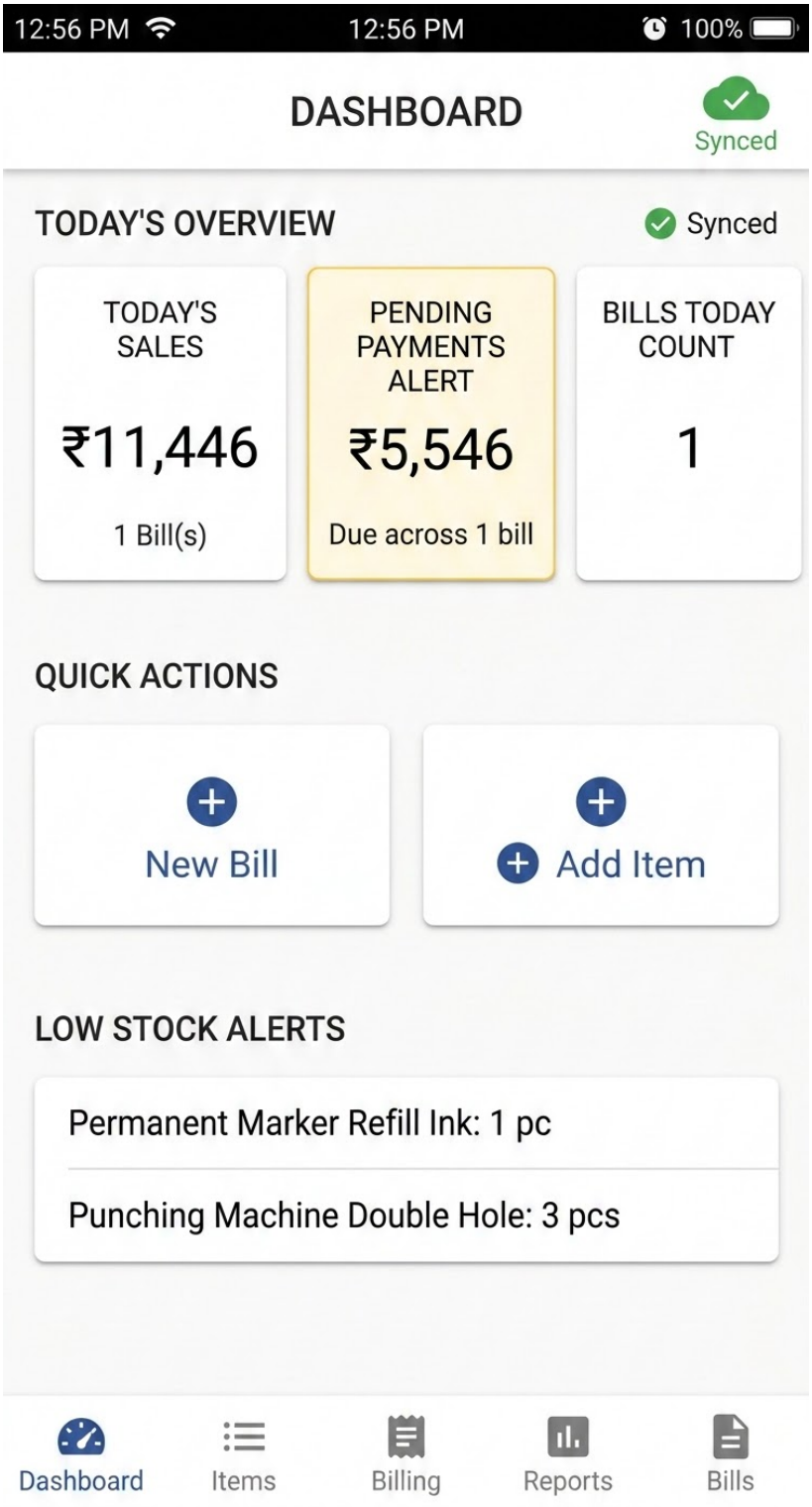
The POS module allows sales staff to search stationery products by keyword, category pill filter, SKU barcode, or godown storage location. Items tap directly into the active bill cart. Quantities are directly editable via numeric keypad input, overriding slow stepper controls.

Toggling the GST Invoice switch updates line items dynamically, displaying per-line tax rates and calculating live totals. When payment is finalized, local SQLite records are written instantly, and a temporary offline bill placeholder is generated for display in Bill History.

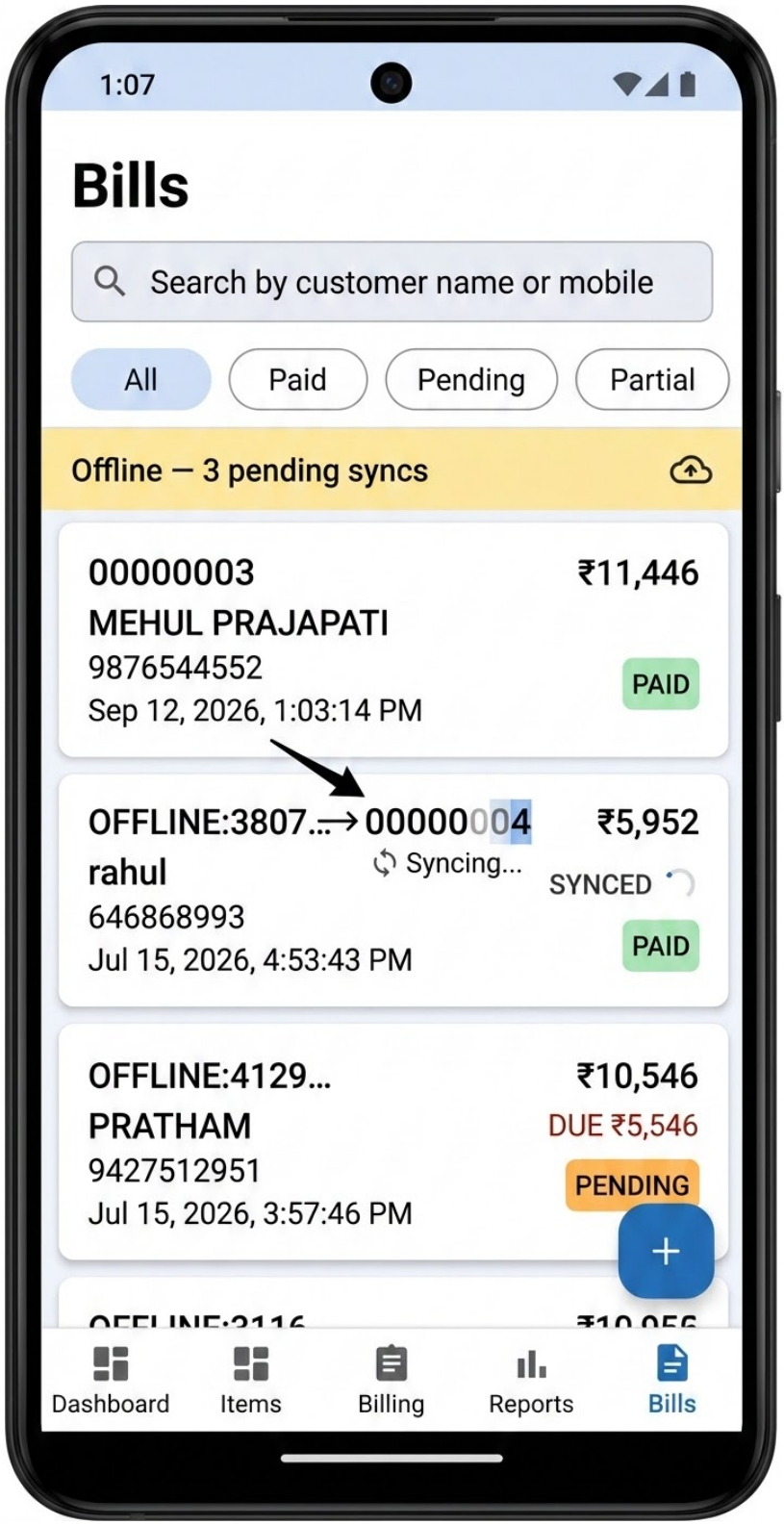
6.3 Offline Sync & Backend Data Flow

When network connectivity returns, `SyncService` drains queued POST/PUT operations in strict chronological order. Upon receiving a new bill payload, the Node.js Express backend inserts the transaction into MySQL 8, triggering an auto-incrementing sequential invoice counter.

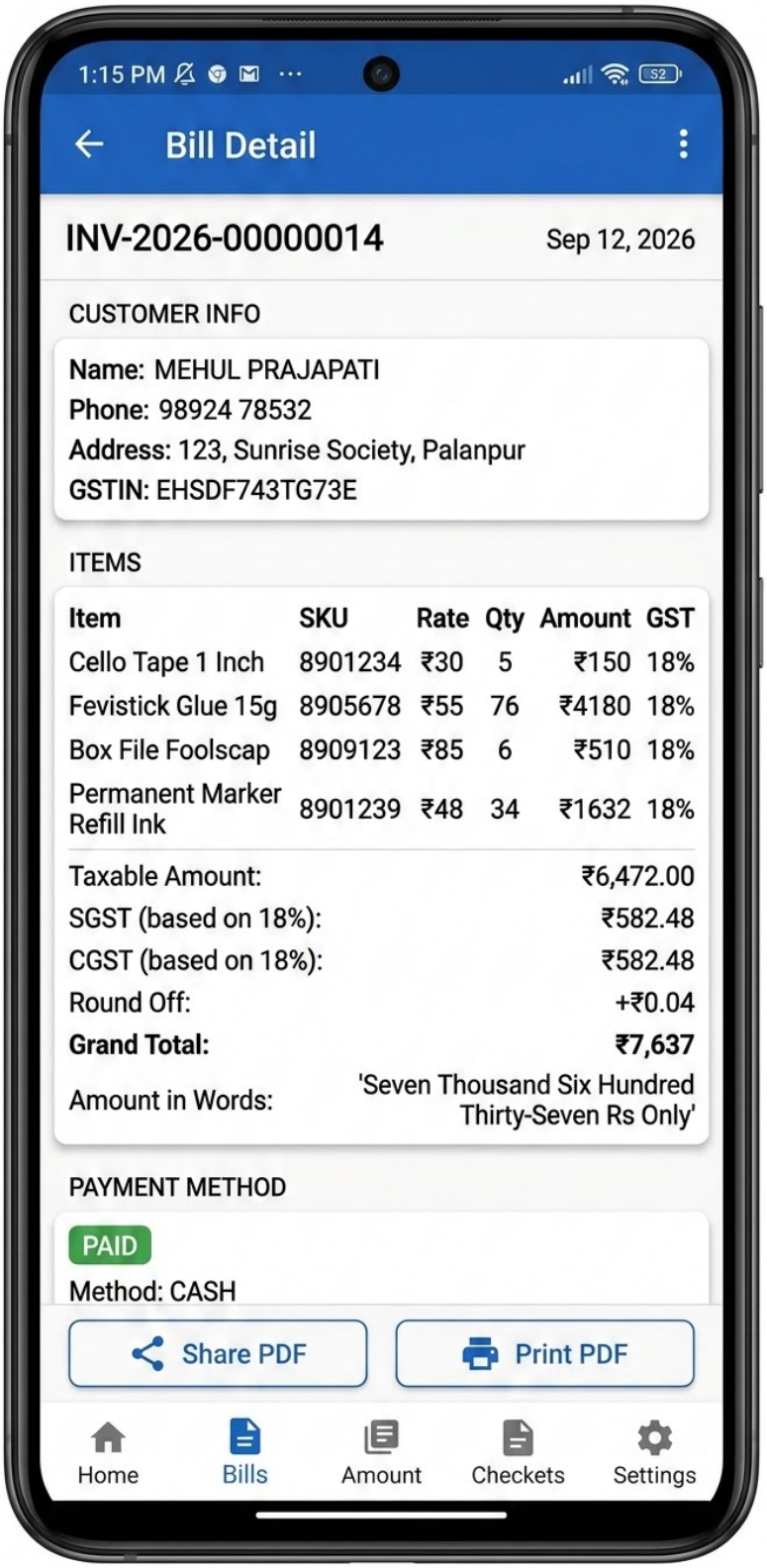
The backend formats this counter into a official 8-digit sequential invoice number (`00000001`, `00000002`) and returns it in the API response. The client application receives the confirmation, updates local SQLite, and seamlessly swaps the offline placeholder tag for the permanent invoice number in the UI without interrupting the user.



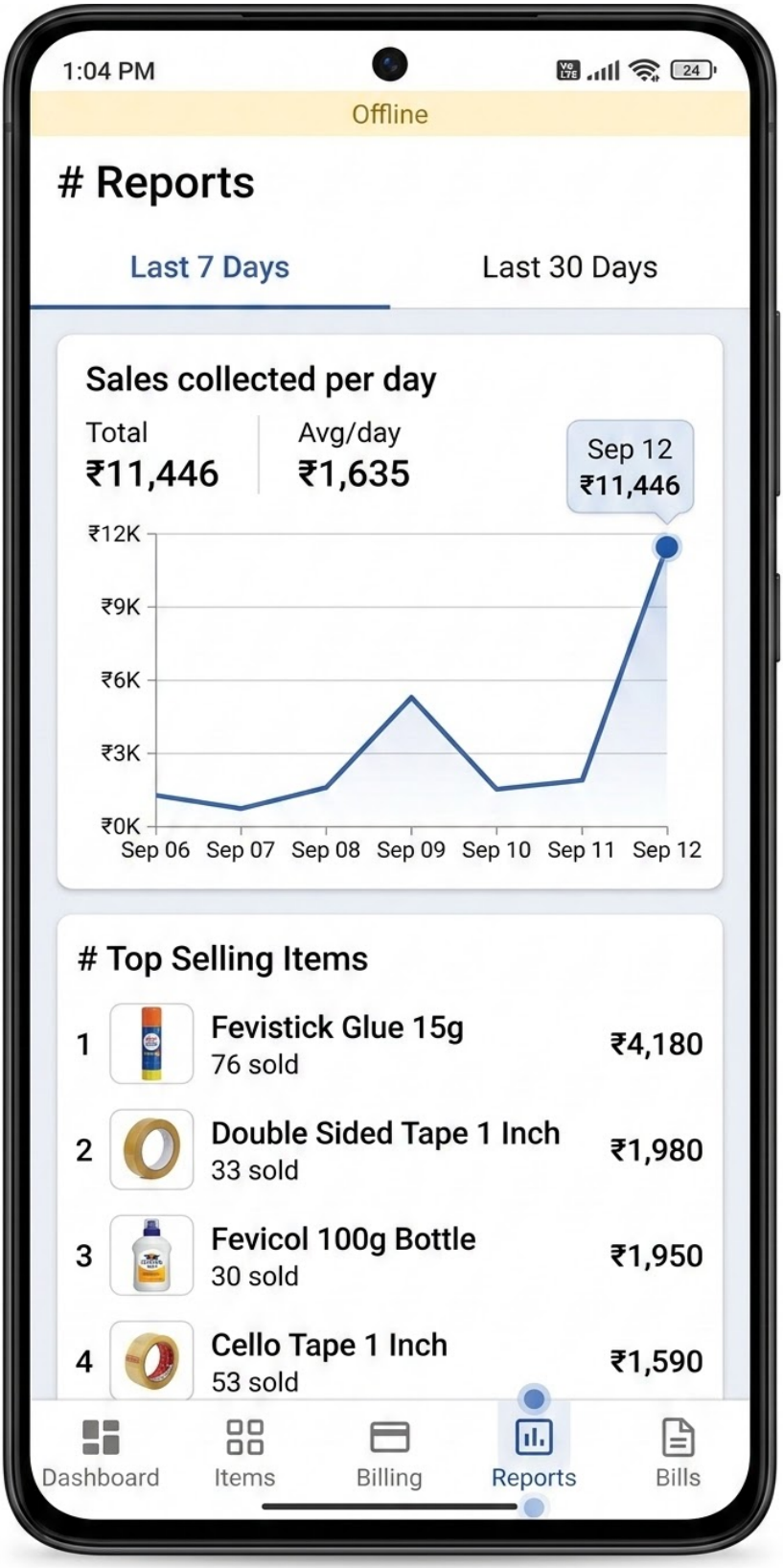
(Fig.6.1 – Ionic Tab Navigation & Dashboard View)



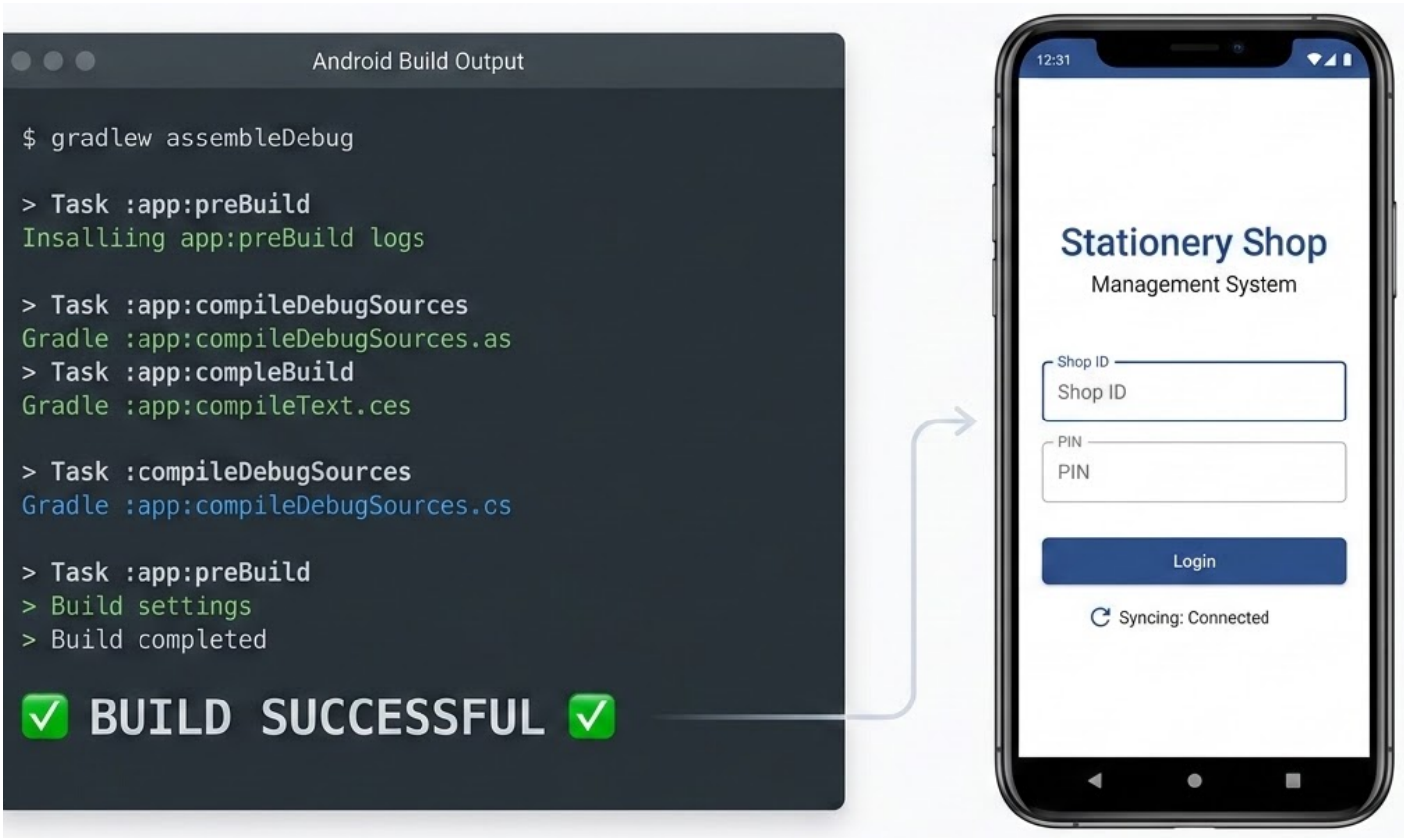
(Fig.6.2 – Bill History & 8-Digit Invoice Swapping Execution)



(Fig.6.3 – Bordered A4 GST Tax Invoice PDF Generated via pdfmake)



(Fig.6.4 – Interactive Sales Analytics Chart & Top Products View)



(Fig.6.5 – Native Android Build Output & Device Deployment)

CHAPTER 7: CONCLUSION & FUTURE SCOPE

7.1 Summary of Achievements

The development of the **Stationery Shop Management Application** successfully demonstrates the deployment of a production-grade, offline-first point-of-sale and inventory system tailored for commercial retail environments. By combining Ionic 8, Angular 20 standalone components, Capacitor 8, SQLite local storage, and a Node.js/MySQL backend, the system eliminates reliance on continuous cloud connectivity.

Key achievements include zero-downtime offline billing, hardware-accelerated camera barcode scanning, complete Indian statutory GST compliance (SGST, CGST, IGST, nearest rupee rounding, Indian number-to-words conversion), automated A4 PDF invoice creation, linear background queue synchronization, and official 8-digit sequential invoice numbering.

The application fulfilled all core technical objectives while providing hands-on experience in cross-platform mobile engineering, local database architecture, tax logic implementation, and native Android deployment pipelines.

7.2 Future Enhancements

While the current version provides comprehensive billing and inventory capabilities, several future extensions can further expand system performance:

1. **Conflict Resolution Engine**: Introducing Operational Transform or CRDT-based multi-device concurrent conflict resolution for multi-register store setups.

2. **Thermal Bluetooth Printer Integration**: Binding ESC/POS direct Bluetooth thermal printing plugins for rapid 2-inch and 3-inch receipt printing.

3. **Automated WhatsApp Invoice Dispatch**: Integrating WhatsApp Business API to send digital PDF tax invoices directly to customer mobile numbers upon checkout.

4. **Cloud Backup & Multi-Store Analytics**: Extending the Express backend to manage centralized inventory across multiple retail chain branches with automated cloud database backups.

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